

IrishAV-Deepfake Baseline Evaluation Results

Simple summary report prepared for repository upload. Source: companion baseline manuscript currently under review.

1. Dataset Used

Subset	Real samples	Synthetic samples	Total samples	Purpose
Audio-only	250	250	500	Audio deepfake detection benchmark
Audio-video/video	246	246	492	Video and multimodal deepfake detection benchmark

2. Evaluation Setup

Item	Description
File-level validation	File signature, file size, format, and metadata checks used as a lightweight file-level gate before AI detection.
Models evaluated	MobileNetV2, ResNet18, ResNet50, EfficientNet-B0, EfficientNet-B3, ViT-B16.
Feature-extraction setting	Pretrained deep learning models used as feature extractors; extracted features classified with classical ML classifiers.
Transfer-learning setting	Pretrained models fine-tuned for binary real/fake classification.
Audio preprocessing	Mel-spectrogram extraction at 16 kHz, 128 mel bands, hop length 160, FFT size 400, frequency range 20-7600 Hz.
Video preprocessing	Frames resized to 224 x 224 pixels and processed sequentially.
Metrics	Accuracy, balanced accuracy, precision, recall, F1 score, ROC AUC, confusion matrix, and runtime where applicable.

3. Feature-Extraction Baselines: Audio

Top-performing representative configuration for each feature extractor.

Feature extractor	ML classifier	Accuracy	Balanced acc.	ROC AUC	F1	Time (s)
MobileNetV2	LinearSVC	0.98 +/- 0.01	0.98 +/- 0.01	0.98 +/- 0.01	0.98 +/- 0.01	0.11
ResNet18	Perceptron	0.99 +/- 0.01	0.99 +/- 0.01	0.99 +/- 0.01	0.99 +/- 0.01	0.02
EfficientNet-B0	RandomForestClassifier	0.97 +/- 0.01	0.97 +/- 0.02	0.97 +/- 0.01	0.97 +/- 0.02	0.87
ViT-B16	RandomForestClassifier	0.99 +/- 0.01	0.99 +/- 0.01	0.99 +/- 0.01	0.99 +/- 0.01	0.62

4. Feature-Extraction Baselines: Video

Top-performing representative configuration for each feature extractor.

Feature extractor	ML classifier	Accuracy	Balanced acc.	ROC AUC	F1	Time (s)
MobileNetV2	LogisticRegression	0.96 +/- 0.01	0.96 +/- 0.03	0.96 +/- 0.05	0.96 +/- 0.02	0.27
ResNet18	CalibratedClassifierCV	0.96 +/- 0.03	0.96 +/- 0.02	0.96 +/- 0.02	0.96 +/- 0.03	0.09
ResNet50	LogisticRegression	0.96 +/- 0.01	0.96 +/- 0.02	0.96 +/- 0.03	0.96 +/- 0.01	0.16
EfficientNet-B0	RidgeClassifierCV	0.96 +/- 0.01	0.96 +/- 0.02	0.96 +/- 0.03	0.96 +/- 0.01	0.16
EfficientNet-B3	RidgeClassifierCV	0.96 +/- 0.01	0.96 +/- 0.03	0.96 +/- 0.02	0.96 +/- 0.03	0.16
ViT-B16	RidgeClassifierCV	0.97 +/- 0.02	0.97 +/- 0.03	0.97 +/- 0.03	0.97 +/- 0.02	0.10

5. Transfer Learning Results: Audio

Model	Accuracy	Precision	Recall	F1 score	ROC AUC
MobileNetV2	0.96 +/- 0.02	0.96 +/- 0.02	0.96 +/- 0.03	0.96 +/- 0.02	0.9868 +/- 0.0012
ResNet18	0.97 +/- 0.01	0.96 +/- 0.02	0.98 +/- 0.02	0.9703 +/- 0.0017	0.9984 +/- 0.0005
EfficientNet-B0	0.98 +/- 0.01	0.96 +/- 0.01	1.00 +/- 0.00	0.9804 +/- 0.0011	0.9984 +/- 0.0004
ViT-B16	0.98 +/- 0.01	0.98 +/- 0.02	0.98 +/- 0.02	0.9800 +/- 0.0013	0.9984 +/- 0.0003

6. Transfer Learning Results: Video

Model	Accuracy	Precision	Recall	F1 score	ROC AUC
MobileNetV2	0.99 +/- 0.01	1.00 +/- 0.00	0.98 +/- 0.01	0.9899 +/- 0.0004	0.9996 +/- 0.0001
ResNet18	0.97 +/- 0.02	0.96 +/- 0.03	0.98 +/- 0.01	0.9703 +/- 0.0020	0.9788 +/- 0.0008
ResNet50	0.99 +/- 0.01	0.98 +/- 0.02	1.00 +/- 0.00	0.9901 +/- 0.0003	1.0000 +/- 0.0000
EfficientNet-B0	0.97 +/- 0.02	0.96 +/- 0.02	0.98 +/- 0.01	0.9703 +/- 0.0032	0.9952 +/- 0.0004
ViT-B16	0.98 +/- 0.01	0.98 +/- 0.02	0.98 +/- 0.01	0.9800 +/- 0.0016	0.9990 +/- 0.0002
EfficientNet-B3	0.98 +/- 0.01	0.98 +/- 0.02	0.98 +/- 0.01	0.9800 +/- 0.0014	0.9980 +/- 0.0005

7. Short Result Summary

The benchmark shows that the IrishAV-Deepfake dataset supports both feature-extraction baselines and transfer-learning evaluation. In the transfer-learning setting, EfficientNet-B0 and ViT-B16 achieved 0.98 accuracy for audio detection. For video detection, MobileNetV2 and ResNet50 achieved 0.99 accuracy, with ResNet50 reaching ROC AUC of 1.0000 on the reported split. These results provide model-based evidence of dataset utility beyond file-level integrity validation.

8. Repository Link

Baseline code and report can be hosted at: https://github.com/kmr2017/IrishDeepFake_Detection